Stock-Based Compensation Expense

The Company recognized stock-based compensation expense as follows (in thousands):

	Year Ended December 31,		
	2019	2018	2017
Cost of revenue	$ 2,409	$ 1,888	$ 1,654
Research and development	19,584	15,990	14,816
Selling, general and administrative	56,706	42,729	36,147
Total stock-based compensation expense	$ 78,699	$ 60,607	$ 52,617
Tax benefit related to stock-based compensation	$ 2,754	$ 4,383	$ 5,054

RSUs

The Company's RSUs include time-based RSUs, RSUs with performance conditions ("PSUs"), RSUs with market conditions ("MSUs"), and RSUs with both market and performance conditions ("MPSUs"). Vesting of awards with performance conditions or market conditions is subject to the achievement of pre-determined performance goals and the approval of such achievement by the Compensation Committee of the Board of Directors (the "Compensation Committee"). All awards include service conditions which require continued employment with the Company.

A summary of RSU activity is presented in the table below (in thousands, except per-share amounts):

	Time-Based RSUs		PSUs and MPSUs		MSUs		Total	
	Number of Shares	Weighted-Average Grant Date Fair Value Per Share	Number of Shares	Weighted-Average Grant Date Fair Value Per Share	Number of Shares	Weighted-Average Grant Date Fair Value Per Share	Number of Shares	Weighted-Average Grant Date Fair Value Per Share
Outstanding at January 1, 2017	366	$ 51.35	2,284	$ 43.24	1,620	$ 23.57	4,270	$ 36.47
Granted	81	$ 94.25	585(1)	$ 62.72	-	$ -	666	$ 66.56
Vested	(175)	$ 48.35	(597)	$ 41.94	-	$ -	(772)	$ 43.39
Forfeited	(14)	$ 61.80	(6)	$ 49.82	-	$ -	(20)	$ 58.46
Outstanding at December 31, 2017	258	$ 66.30	2,266	$ 48.59	1,620	$ 23.57	4,144	$ 39.91
Granted	133	$ 114.36	630(1)	$ 85.06	600	$ 68.48	1,363	$ 80.62
Vested	(136)	$ 60.23	(717)	$ 41.08	-	$ -	(853)	$ 44.13
Forfeited	(15)	$ 82.20	(5)	$ 63.16	(1)	$ 68.48	(21)	$ 76.92
Outstanding at December 31, 2018	240	$ 95.38	2,174	$ 61.61	2,219	$ 35.69	4,633	$ 50.94
Granted	52	$ 141.32	512(1)	$ 99.88	-	$ -	564	$ 103.68
Vested	(103)	$ 81.53	(656)	$ 53.72	(324)	$ 23.57	(1,083)	$ 47.34
Forfeited	(9)	$ 117.31	(43)	$ 42.72	(9)	$ 68.48	(61)	$ 57.01
Outstanding at December 31, 2019	180	$ 115.45	1,987	$ 74.50	1,886	$ 37.63	4,053	$ 59.16

(1) Amount reflects the number of awards that may ultimately be earned based on management's probability assessment of the achievement of performance conditions at each reporting period.

The intrinsic value related to vested RSUs was $138.3 million, $90.0 million and $74.0 million for the years ended December 31, 2019, 2018 and 2017, respectively. As of December 31, 2019, the total intrinsic value of all outstanding RSUs was $679.5 million, based on the closing stock price of $178.02. As of December 31, 2019, unamortized compensation expense related to all outstanding RSUs was $100.1 million with a weighted-average remaining recognition period of approximately three years.

Cash proceeds from vested PSUs with a purchase price totaled $16.6 million and $10.6 million for the years ended December 31, 2019 and 2018, respectively. There were no proceeds for the year ended December 31, 2017.

Time-Based RSUs

For the years ended December 31, 2019, 2018 and 2017, the Compensation Committee granted 52,000, 133,000, and 81,000 RSUs, respectively, with service conditions to non-executive employees and non-employee directors. The RSUs generally vest over four years for employees and one year for directors, subject to continued service with the Company.

PSUs and MPSUs

2019 PSUs:

In February 2019, the Compensation Committee granted 151,000 PSUs to the executive officers, which represent a target number of shares to be earned based on the Company's average two-year (2019 and 2020) revenue growth rate compared against the analog industry's average two-year revenue growth rate as published by the Semiconductor Industry Association ("2019 Executive PSUs"). The maximum number of shares that an executive officer can earn is 300% of the target number of the 2019 Executive PSUs. 50% of the 2019 Executive PSUs will vest in the first quarter of 2021 if the pre-determined performance goals are met during the performance period. The remaining 2019 Executive PSUs will vest over the following two years on a quarterly basis. Assuming the achievement of the highest level of performance goals, the total stock-based compensation cost for the 2019 Executive PSUs is $46.6 million.